A Short Note on Stars

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Abstract

Stars are giant balls of hot gas held together by their own gravity. In this note we describe how stars produce energy by nuclear fusion, how their lifetime depends on their mass, and how their final state (white dwarf, neutron star or black hole) is set by the mass they have left at the end of their lives.

1 Introduction

The Sun is an ordinary, medium-sized star and the closest star to the Earth. Light from the Sun takes about 8 minutes to reach us. Stars produce their energy by nuclear fusion in their cores, where hydrogen nuclei fuse into helium and release energy [1].

2 Stellar evolution

A star spends most of its life on the main sequence, where the pressure of the hot gas balances gravity. Massive stars live for a much shorter time than small stars, because they use up their fuel faster. Table 1 lists three typical stars.

Table 1: Typical main-sequence stars. Lifetimes are order-of-magnitude estimates.

Star type	Mass (Sun = 1)	Surface temperature (K)	Lifetime (years)
Red dwarf	0.2	3 000	10^{12}
Sun-like	1	5 800	10^{10}
Blue giant	20	30 000	10^{7}

When a star runs out of hydrogen in its core it leaves the main sequence. Its final state depends on its mass: stars like the Sun end as white dwarfs, while the most massive stars collapse into neutron stars or black holes.

References

[1] B. W. Carroll and D. A. Ostlie, *An Introduction to Modern Astrophysics*, 2nd ed., Cambridge University Press, 2017.